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Transport Costs and the Dynamic Stability of Regional Agglomeration

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Transport infrastructure and where people locate

- Transport infrastructure shapes *where* people live and work.
- Highways and high-speed rail are **large**, **long-lived**, and **hard to reverse**.
- A central question for project appraisal **from the long-run perspective**:
 - Does better connectivity **concentrate** activity in a few big hubs?
 - Or does it **spread** activity toward smaller, peripheral regions?
- This matters directly for whether infrastructure can **revitalize lagging regions**, a common policy goal.

“Will this new line help the periphery, or just pull more activity into the capital?”

Theory gives two opposite answers

A lower transport cost can work in *two opposite directions*:

(1) It **weakens the advantage of being big**.

Distance does not matter anymore $\Rightarrow$ some activities can **disperse**

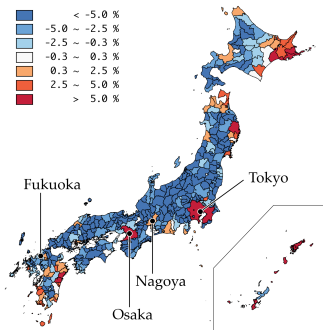
(2) It **expands the market reach of big hubs**

Large places sell to everyone $\Rightarrow$ some activities can **concentrate** further.

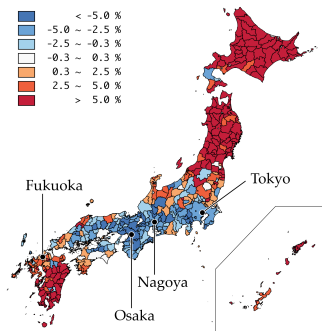
- Which effect wins is not obvious, and it can flip the *sign* of a counterfactual.
- Before trusting a number, we need to understand the **mechanics** of the model.

Same shock, opposite predictions

A concrete example: *removing highways* (Sugimoto *et al.*, 2025). (Transport costs $\uparrow$)



more concentration in the core cities



spreading to the periphery

- Same data, similar model, same shock. But the *sign* of the answer reverses.

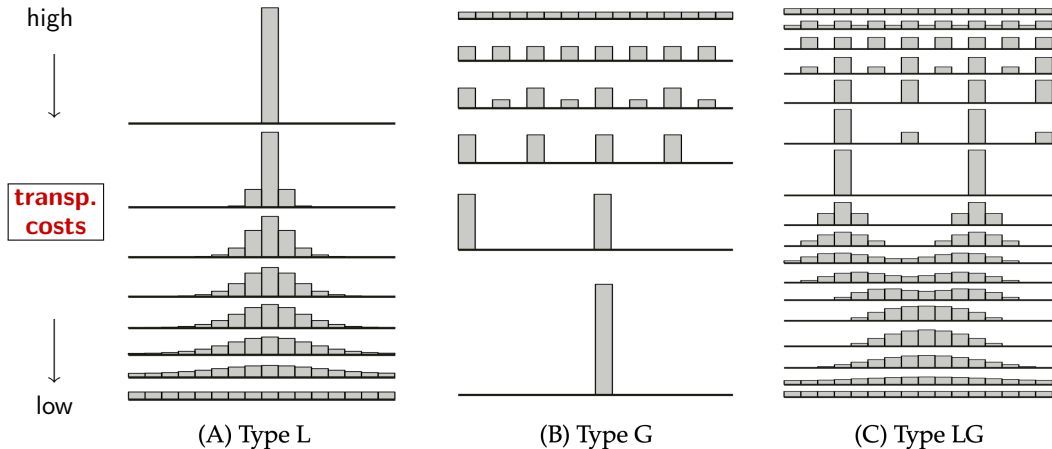
Two kinds of dispersion forces (Akamatsu *et al.*, 2026)

The key is *what kind of dispersion force* the model contains.

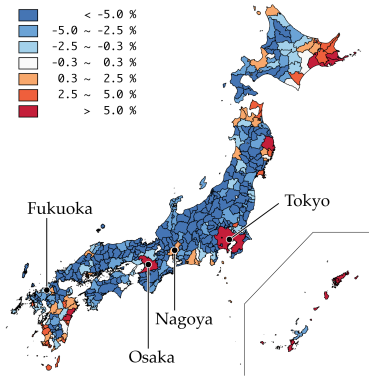
- **Local** dispersion force: crowding *within* a place.
e.g., land rents, commuting costs (Helpman, 1998).
- **Global** dispersion force: competition *across* places.
e.g., firms competing for the same customers and immobile factors (Krugman, 1991).

Type L (local)	Type G (global)	Type LG
Helpman (1998), Redding & Sturm (2008), Allen & Arkolakis (2014)	Krugman (1991), Puga (1999), Harris & Wilson (1978)	Tabuchi (1998), Kucheryavyi <i>et al.</i> (2024), Sugimoto <i>et al.</i> (2025)

Evolution of spatial patterns in a stylized economy

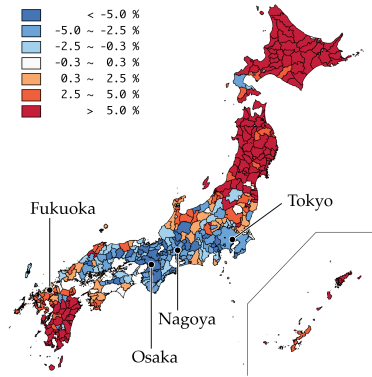


Same shock, opposite predictions



Local dispersion force only (Type L)

Majority of conventional regional QSMs



Global dispersion force added (Type LG)

Sugimoto *et al.* (2025)

But the real world is dynamic

So far the story is *static*. In reality, space evolves over time:

- Population changes through **generations** and **forward-looking migration**.
- Today's productivity and amenities depend on the **past** (infrastructure, human capital, reputation).
- Infrastructure effects *unfold over decades*.
- Many policies aim at **balanced, multi-center (polycentric) development**.
- They implicitly assume polycentric structures are *stable in the long run*.
- *Is that true once we model the dynamics?* This is the gap we address.

This study

- We take a seminal *dynamic* quantitative spatial model and ask what spatial patterns it produces **on its own**.
- Specific subject: **Allen & Donaldson (2020) [AD]** model
(“Persistence and Path Dependence in the Spatial Economy”)
 - Clean, tractable, with sound microfoundations.
 - A dynamic cousin of the static Allen & Arkolakis (2014) model.
- Two questions:
 - Q1. Can this dynamic economy sustain a *polycentric structure*?
 - Q2. Or do the forces ultimately push toward a *single core*?
- Method: study agglomeration as *instability of the symmetric (even) state*.

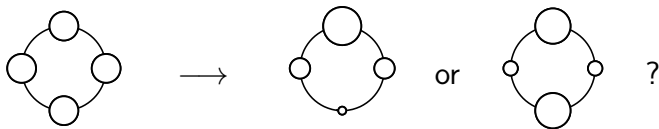
The (symmetric) AD framework

We use the most symmetric version, so geography adds no bias.

- N locations with symmetric characteristics.
- Iceberg *trade frictions* between locations $\{\tau_{ij}\}$, $\tau_{ij} \geq 1$.
- Iceberg *migration frictions* between locations $\{\mu_{ij}\}$, $\mu_{ij} \geq 1$.
- Population distribution $\mathbf{L}_t = (L_{i,t})_{i=1}^N$ at time $t \in \{0, 1, 2, \dots\}$.
- Competitive Armington trade, with *local* externalities that link generations:
 - Amenity: $u_i = L_{i,t}^{\beta_1} \cdot L_{i,t-1}^{\beta_2}$ ($\beta_1 < 0$, $\beta_2 > 0$)
 - Productivity: $a_i = L_{i,t}^{\alpha_1} \cdot L_{i,t-1}^{\alpha_2}$ ($\alpha_1 > 0$, $\alpha_2 > 0$)
- Trade + migration equilibrium gives discrete-time dynamics: $\mathbf{L}_t = \mathbf{F}(\mathbf{L}_{t-1})$.

A four-location ring: a minimal testbed

- With symmetric geography, we can study the *even steady state* $\bar{L} = (\frac{1}{N}, \dots, \frac{1}{N})$, since $\bar{L} = F(\bar{L})$.
- If $\bar{L}$ becomes *unstable*, some form of endogenous agglomeration appears.
- Four locations on a circle is the simplest case that still distinguishes patterns:

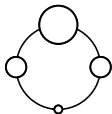


- Even $\rightarrow$ **mono-centric** (one city) or **poly-centric** (two cities)?
 - Static **Type L** (local disp. force only): only mono-centric pattern.
 - Static **Type G** (global disp. force only): poly-centric pattern is possible.

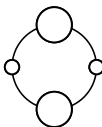
Agglomeration as instability of the even state

- We can separately study each **mode** k .

$k = 1$: **mono-centric** direction



$k = 2$: **poly-centric** direction



- Each mode has a scalar “**net agglomeration force**”

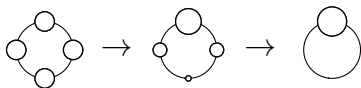
$$f_k \approx \frac{\text{agglomeration force}}{\text{dispersion force (inertia)}}$$

- $\bar{L}$ is stable when $|f_k| < 1$:

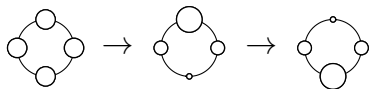
$$\bar{L} + \varepsilon_t \mapsto \bar{L} + f_k \varepsilon_{t+1}$$

Agglomeration as instability of the even state

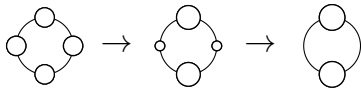
- Discrete time $\Rightarrow$ the **sign** of f_k matters too: $f_k > 0$ *monotone*, $f_k < 0$ *oscillating*.
- Four possible transition paths:



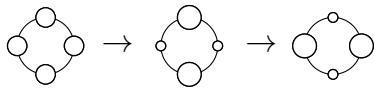
M1 : monotone, monocentric ($k = 1$)



O1 : oscillatory, monocentric ($k = 1$)



M2 : monotone, poly-centric ($k = 2$)



O2 : oscillatory, poly-centric ($k = 2$)

Breaking the even state: sum vs. difference

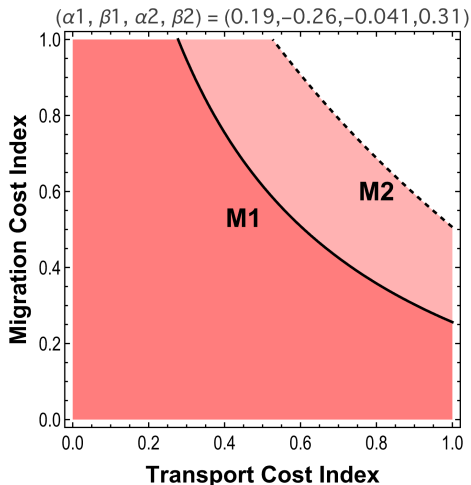
The thresholds where the even state breaks have a clean “*sum vs. difference*” form:

$$\begin{aligned} \text{monotone } (f_k = +1) : \quad & (\alpha_1 + \alpha_2)(1 - \rho_k) + (\beta_1 + \beta_2) - \rho_k = \frac{1 - \delta\lambda_k}{\theta(1 + \lambda_k)} \\ \text{oscillating } (f_k = -1) : \quad & (\alpha_1 - \alpha_2)(1 - \rho_k) + (\beta_1 - \beta_2) - \rho_k = \frac{1 + \delta\lambda_k}{\theta(1 - \lambda_k)} \end{aligned} \quad (k = 1, 2)$$

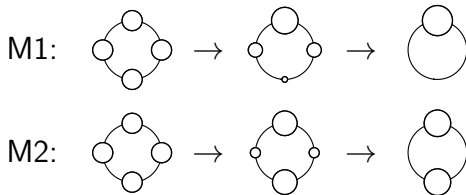
- *Monotone* agglomeration depends only on the **sum** of externalities (t and $t - 1$). Then mode $k = 1$ always breaks first $\Rightarrow$ **the pattern is always mono-centric**.
- This mirrors the static Allen & Arkolakis (2014): AD behaves like **Type L**.
- Poly-centric outcomes need *oscillation*, which depends on the **difference** of externalities. A genuinely *dynamic* channel, but empirically demanding.

Result 1: Monocentric agglomeration under US estimates

Stability region in the cost plane (dark = stable), at AD's estimated elasticities.

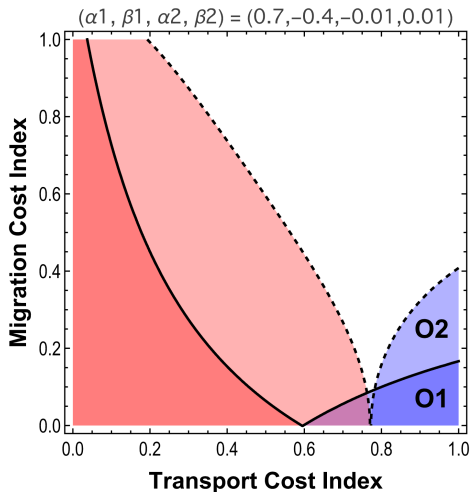


- Transition path should look like **monotone and mono-centric**.



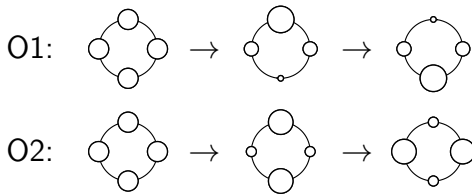
- $\approx$ **Type L**. AD predicts dispersion when transport costs go down.

Result 2: Strong agglomeration forces can bring oscillation



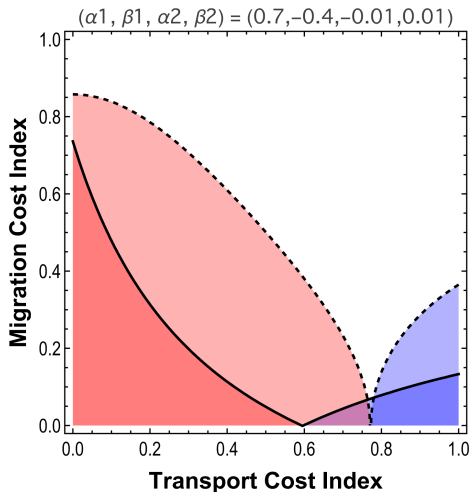
Strong agglomeration, no foresight ($\delta = 0$)

- If the current-period agglomeration forces are extreme, other symmetry breaks appear: **oscillating** (O1, O2) paths.



- In AD, oscillation is the only route to polycentricity. But it needs empirically demanding parameter values, and unlikely in practice.

Result 3: Forward-looking migration



With foresight ($\delta = 0.3$)

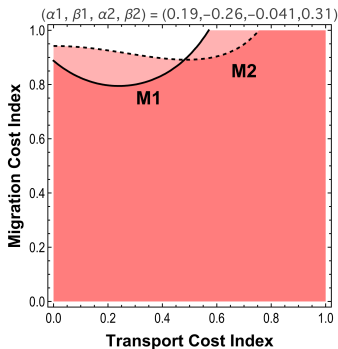
- In the full AD model, agents may anticipate payoffs of future generation.
- The stable region *shrinks*, but the qualitative story is unchanged.
- Migration frictions act as an *extra* push toward *mono-centric* outcomes.
- Agents' foresight does **not** buy you polycentricity but it strengthens concentration.

Why no poly-centricity? A minimal extension

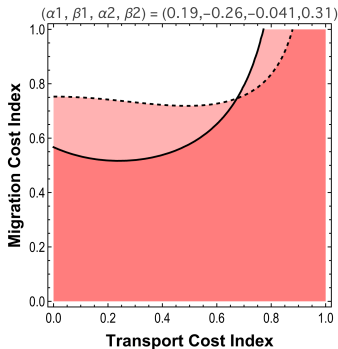
- AD has *no global dispersion force*, so poly-centric patterns are never stable.
- Minimal extension: a constant *immobile demand* $\bar{R} > 0$ in each region.
 - Reduced form for government spending, resource income, pensions, etc.
 - = Krugman's immobile farmers (cf. Kucheryavyy *et al.* (2024)).
 - Mobile demand share $s = \frac{wL}{wL + \bar{R}} \in (0, 1]$.
 - In general, $s < 1$. Back-of-envelope estimate for Japan is $s \approx 0.7$.
- With $\bar{R} > 0$:
 - *high* transport cost $\rightarrow$ poly-centric agglomeration path (M2);
 - *low* transport cost $\rightarrow$ mono-centric agglomeration path (M1).

Result: the extended model produces poly-centricity

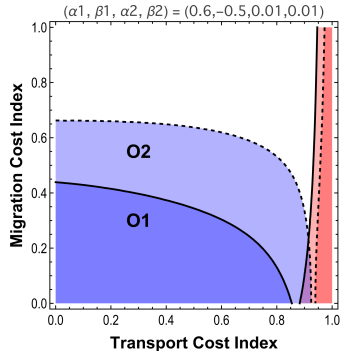
Stability regions with immobile demand (mobile demand share $s = 0.5$).



AD estimates, $\delta = 0$



With foresight, $\delta = 0.3$



Strong agglomeration

Some interpretations on “immobile demand”

- Where does immobile demand $\bar{R}$ come from? A natural source: the *old generation*.
 - They have finished migrating, still consume, but their income (pensions) does *not* depend on where production happens this period.
- So aging raises the immobile-demand share (and lowers the mobile share s).
- This *supplies a global dispersion force*
⇒ can move the economy from **Type L** toward **Type LG**.
- Demographic aging may make poly-centric structures more sustainable?

Summary

- In theory, better connectivity does **not** automatically promote balanced development. In a Type-LG economy it tends to *deepen* concentration in existing hubs.
- The sign of a transport counterfactual tend to depend on the **type of dispersion force** in the model in *both static and dynamic settings*.

	Type L	Type G	Type LG
Static	Helpman (1998), Allen & Arkolakis (2014)	Krugman (1991), Puga (1999)	Tabuchi (1998), Kucheryavyi et al. (2024)
Dynamic	Allen & Donaldson (2020)	???	Allen & Donaldson + $\bar{R}$

- In both AA and AD, endogenous agglomeration is *structurally mono-centric*.
- Stylized geographies remain powerful for reading the mechanics of a spatial model.

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